# Bahram Afsharmanesh

Curriculum Vitae

University of Georgia
Athens, Georgia, USA
Google Scholar | LinkedIn

bahram.afsharmanesh@uga.edu
bafsharmanesh.github.io

## Research Interests

High-performance computing and machine learning systems; heterogeneous GPU scheduling and distributed training; GPU kernels and collective communication; communication–computation overlap, quantization, and performance modeling for large language model inference.

## Education

**University of Georgia** 2023–Present
Ph.D. in Computer Science, in progress GPA: 3.95/4.00
Advisor: Prof. Gagan Agrawal.

**Sharif University of Technology** 2012–2014
M.Sc. in Mechanical Engineering GPA: 3.81/4.00

**Iran University of Science and Technology** 2008–2012
B.Sc. in Mechanical Engineering GPA: 3.76/4.00

## Selected Publications

[1] **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, AmirAli Mirian, and Gagan Agrawal. **PhaseAR: Schedule-Aware AllReduce for Tensor-Parallel LLM Decoding**, Under Review.

[2] **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, AmirAli Mirian, and Gagan Agrawal. **LiGO: Low-Bit GEMM–AllReduce Overlap for Tensor-Parallel LLM Inference.**, Under Review.

[3] AmirAli Mirian, **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, Wei Niu, and Gagan Agrawal. **Fast Sparse Linear Algebra for LLMs with Contextual Sparsity.**, Under Review.

[4] **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, AmirAli Mirian, and Gagan Agrawal. **HeteroSched: Co-Optimizing Scheduling and Parallelization for Deep Learning Workloads for Heterogeneous GPU Clusters.** *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2026.

[5] Md Musfiqur Rahman Sanim, Zhihao Shu, **Bahram Afsharmanesh**, AmirAli Mirian, Wei Niu, and Gagan Agrawal. **Mask-Aware Execution for Efficient JEPA Training.** *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2026.

[6] Md Musfiqur Rahman Sanim, Zhihao Shu, **Bahram Afsharmanesh**, AmirAli Mirian, Jiexiong Guan, Wei Niu, Bin Ren, and Gagan Agrawal. **Optimizing 3D Gaussian Splattering for Mobile GPUs.** *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2025.

## Selected Presentations

**HeteroSched** 2026
International Conference on Parallel Architectures and Compilation Techniques (PACT), Chicago, IL.

## Professional Service

**Artifact Evaluation Committee Member** 2027
International Symposium on Code Generation and Optimization (CGO).

**Technical Paper Reviewer** 2027
IEEE International Symposium on Circuits and Systems (ISCAS).

**Junior Program Committee Member** 2027
ACM International Conference on Supercomputing (ICS).

**Artifact Evaluation Committee Member** 2027
Symposium on Principles and Practice of Parallel Programming (PPoPP).
**Artifact Evaluation Committee Member** 2026
International Conference on Parallel Architectures and Compilation Techniques (PACT).

## Honors and Awards

**Kaggle Silver Medal** 2023
AMP Parkinson's Disease Progression Prediction competition; ranked 47th of 1,805 (top 3%).
**10th Place, National Student Scientific Olympiad** 2012
Mechanical Engineering, Iran.
**10th Place, National M.Sc. Entrance Examination** 2012
Mechanical Engineering, Iran.
**Top 0.2%, Iranian National University Entrance Examination** 2008
Ranked 421st among over 200,000 participants nationwide.

## Research Experience

**Graduate Research Assistant** Aug. 2023–Present
*University of Georgia; advisor: Prof. Gagan Agrawal*

- Designed HeteroSched to jointly optimize scheduling and parallelization for heterogeneous GPU clusters; evaluated with performance simulation and physical experiments.
- Developed CUDA/C++ and MSCCL++ collective-communication kernels, fine-grained communication–computation overlap, and quantized AllReduce for distributed LLM inference over PCIe and NVLink.
- Built microbenchmarks and analytical performance models; used Nsight Systems and Nsight Compute to study synchronization, memory bandwidth, kernel occupancy, and end-to-end performance.

**Research Assistant** May 2020–June 2021
*Amirkabir University of Technology*

- Investigated machine learning as a function approximator for numerical PDE solvers; developed MATLAB applications for mechanical-system analysis under static, dynamic, and thermal loads.

**Research Assistant** Sept. 2012–Sept. 2015
*Sharif University of Technology*

- Developed PDE solvers in MATLAB and molecular dynamics algorithms in Fortran; used genetic algorithms to calibrate numerical models against molecular dynamics simulations.

**Undergraduate Researcher** Sept. 2010–Sept. 2012
*Iran University of Science and Technology; advisor: Prof. S. M. Hasheminejad*

- Solved PDEs for laminated plates using MATLAB and Mathematica.

## Industry Experience

**Data Scientist** Aug. 2021–Dec. 2022
*Baroline*

- Built spam-detection and named-entity-recognition models with PyTorch, BERT, and scikit-learn; deployed inference APIs using FastAPI, Docker, and Kafka, reducing response latency by 25%.
- Developed data pipelines and warehouses with Airflow and PostgreSQL, and analytics dashboards with Pandas, Streamlit, Power BI, and Metabase.

**R&D Engineer** Jan. 2016–Apr. 2020
*MAPNA PARS Company*

- Built MATLAB toolboxes for system identification and eigenvalue analysis; developed regression models to predict component behavior and accelerate engineering design workflows.

## Technical Skills

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| **Programming** | C, C++, Python, CUDA, Fortran, SQL, MATLAB, Mathematica, Bash |
| **GPU and parallel** | MSCCL++, NCCL, CUTLASS/CuTe, Triton, MPI, OpenMP |
| **ML and data** | PyTorch, JAX, scikit-learn, NumPy, Pandas, FastAPI, Kafka, Airflow |
| **Performance** | Nsight Systems, Nsight Compute, profiling, benchmarking, analytical performance modeling, roofline analysis |
| **Development** | Linux, Git, Docker, Make/CMake, CI/CD, AWS, LaTeX |